

A REAL NON-BOREL DIRICHLET DILATION

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1. SOURCE QUESTION

Kalenda–Spurny [1] ask the following in Question 7.10, after establishing a complex example under the continuum hypothesis where $D1$ is not Borel.

$F^* \circ \phi$ $|\mu|$ -a.e., hence the assertion follows. $\square$

Theorems 7.3 and 7.8 provide partial extension of Fact 7.3 to spaces without constants. However, they do not provide a complete analogue. In fact, the complete analogue fails as witnessed by the following example.

Example 7.9. *Under the continuum hypothesis there is a complex simplicial function space H on a compact space K such that the function $D1$ is not Borel.*

Proof. Let K and H be as in Example 6.17. Using the notation from this example and its proof, we have (by Step 5 of the proof)

$$\delta_{(t,s,0)} = \frac{\overline{sg(t)}}{2} (\varepsilon_{(t,g(t),-1)} + \varepsilon_{(t,g(t),1)}) \text{ for } s, t \in \mathbb{T}.$$

Therefore $D1(t, s, 0) = \overline{sg(t)}$ for $s, t \in \mathbb{T}$. In particular, $D1(t, 1, 0) = \overline{g(t)}$ for $t \in \mathbb{T}$ which is not a Borel function. (Otherwise, g would be a Borel function, so its graph, being an uncountable Borel subset of $\mathbb{T} \times \mathbb{T}$, would contain a Cantor set (by [13] Theorem 13.6)), in contradiction with Lemma 6.18. $\square$

Anyway, the following question remains open.

Question 7.10. *Let H be a simplicial function space not containing constants on a (non-metrizable) compact space K . Let $f : K \rightarrow \mathbb{F}$ be continuous.*

- *Is the function $\tilde{D}f$ Borel?*
- *Assume $\mathbb{F} = \mathbb{R}$. Is the function Df Borel? Is it H -affine?*

The real part of the second bullet asks whether, for a real simplicial function space H not containing constants on a non-metrizable compact space K , the function Df must be Borel for every continuous $f : K \rightarrow \mathbb{R}$. The answer is negative.

2. PROOF INTUITION

The complex counterexample in [1, Example 7.9] encodes a non-Borel function g into the representing boundary measures, so that $D1(t, 1, 0) = g(t)$. The same encoding does not need complex scalars if we only want a real non-Borel $D1$: a two-valued non-Borel function $\varepsilon : [0, 1] \rightarrow \{-1, 1\}$ is enough. The second circle coordinate in the complex example can be replaced by the two-point compact space $\{-1, 1\}$. The Stacey porcupine compactum still lets us split the graph of ε , and the relation $f(t, s, 0) = sf(t, 1, 0)$ forces the unique boundary representative of $(t, s, 0)$ to be the signed average of the two split graph points with coefficient $s\varepsilon(t)$. Thus the

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dilation of the constant function is $D1(t, s, 0) = s\varepsilon(t)$, which is non-Borel on the closed slice $s = 1$.

3. THE COUNTEREXAMPLE

Theorem. *There are a non-metrizable compact space K , a closed real function space $H \subset C(K, \mathbb{R})$ not containing nonzero constant functions, and a continuous function $f : K \rightarrow \mathbb{R}$, such that H is simplicial and the Dirichlet dilation Df is not Borel. One may take $f \equiv 1$.*

Proof. Let $X = [0, 1]$, and choose a non-Borel set $B \subset X$. Define

$$\varepsilon(t) = \begin{cases} 1, & t \in B, \\ -1, & t \notin B. \end{cases}$$

Then $\varepsilon : X \rightarrow \{-1, 1\}$ is not Borel. Let $S = \{-1, 1\}$, $L = X \times S$, and

$$A = \{(t, \varepsilon(t)) : t \in X\} \subset L.$$

Let $K = K_{L,A}$ be the Stacey porcupine compactum recalled in [1, Section 6]: its points are

$$(L \times \{0\}) \cup (A \times \{-1, 1\}),$$

the points in $A \times \{-1, 1\}$ are isolated, and a neighborhood base of $(\ell, 0) \in L \times \{0\}$ is obtained from neighborhoods of ℓ in L after deleting the two split points over ℓ . Stacey's theorem, quoted in [1, Section 6], gives that K is compact Hausdorff. Since A is uncountable and its two split copies are isolated, K is not metrizable.

Write points of K as $(t, s, 0)$ for $(t, s) \in L$, and as $(t, \varepsilon(t), j)$ for $j = \pm 1$ on the split graph. Define H to be the set of all $h \in C(K, \mathbb{R})$ satisfying

$$(1) \quad h(t, s, 0) = s h(t, 1, 0) \quad (t \in X, s \in S),$$

$$(2) \quad h(t, \varepsilon(t), 0) = \frac{1}{2}(h(t, \varepsilon(t), -1) + h(t, \varepsilon(t), 1)) \quad (t \in X).$$

These pointwise linear equations define a closed linear subspace of $C(K, \mathbb{R})$. It contains no nonzero constant functions, since (1) with $s = -1$ forces a constant c to satisfy $c = -c$.

The space H separates points. Indeed, for every $u \in C(X)$ the function

$$U(t, s, 0) = s u(t), \quad U(t, \varepsilon(t), j) = \varepsilon(t) u(t)$$

belongs to H ; such functions separate the base points $(t, s, 0)$. For a fixed $t_0 \in X$, the function q_{t_0} which is 1 at $(t_0, \varepsilon(t_0), 1)$, -1 at $(t_0, \varepsilon(t_0), -1)$, and 0 elsewhere is continuous on the porcupine compactum and belongs to H . These functions separate the two split points over t_0 and separate split points from base points.

We next identify the boundary representatives. For $t \in X$ and $s \in S$ set

$$\delta_{(t,s,0)} = \frac{s\varepsilon(t)}{2} (\varepsilon_{(t,\varepsilon(t),-1)} + \varepsilon_{(t,\varepsilon(t),1)}),$$

and for split graph points set

$$\delta_{(t,\varepsilon(t),j)} = \varepsilon_{(t,\varepsilon(t),j)} \quad (j = \pm 1).$$

The relations (1)–(2) show immediately that these measures represent the corresponding evaluation functionals and have norm one, which is the norm of every evaluation functional because the function $(t, s, 0) \mapsto s$, $(t, \varepsilon(t), j) \mapsto \varepsilon(t)$, belongs to the unit ball of H and has modulus one everywhere.

It remains to see uniqueness among H -boundary measures. This is the real two-point version of the proof of [1, Example 6.17, Step 5]; we include the argument for completeness. The split points are the Choquet boundary points: the functions q_t expose the two split evaluations over $(t, \varepsilon(t))$, while

$$\phi(t, s, 0) = \frac{s\varepsilon(t)}{2} (\phi(t, \varepsilon(t), -1) + \phi(t, \varepsilon(t), 1))$$

shows that base evaluations are not extreme. Hence, inside each finite fiber over t , a maximal positive measure is supported by the two split points.

Let $\mu \in M_x(H)$ be H -boundary. Choose $u \in C(X)$ with $u(t_0) = 1$ and $|u(t)| < 1$ for $t \neq t_0$, and use the corresponding function $U(t, s, 0) = su(t)$, $U(t, \varepsilon(t), j) = \varepsilon(t)u(t)$. If x lies over t_0 , the equality

$$|U(x)| = 1 = \|\mu\|$$

forces $|\mu|$ to be supported by the finite fiber over t_0 . Since μ is boundary, the preceding paragraph removes the two base points in that fiber. Thus μ is supported by

$$\{(t_0, \varepsilon(t_0), -1), (t_0, \varepsilon(t_0), 1)\}.$$

Writing $\mu = a\varepsilon_{(t_0, \varepsilon(t_0), -1)} + b\varepsilon_{(t_0, \varepsilon(t_0), 1)}$, the equations obtained by testing against q_{t_0} and against $(t, s, 0) \mapsto s$ give:

$$-a + b = q_{t_0}(x), \quad \varepsilon(t_0)(a + b) = s_x,$$

where $s_x = \varepsilon(t_0)$ for split points and $s_x = s_0$ for $x = (t_0, s_0, 0)$. These equations give exactly the measures displayed above. Hence H is simplicial.

Now take the continuous function $f \equiv 1$ on K . By the definition of the dilation,

$$D1(t, s, 0) = \int_K 1 d\delta_{(t,s,0)} = s\varepsilon(t).$$

Restricting to the closed slice

$$E = \{(t, 1, 0) : t \in X\} \subset K,$$

which is homeomorphic to X , gives

$$D1(t, 1, 0) = \varepsilon(t).$$

If $D1$ were Borel on K , then its restriction to E would be Borel, and consequently ε would be Borel on X , contradicting the choice of B . Therefore $D1$ is not Borel. $\square$

4. VERIFICATION AND LIMITATIONS

The construction answers only the real Borel part of Question 7.10. It does not address whether $\tilde{D}f$ must be Borel, and it does not settle the remaining universal H -affinity question beyond showing that Df can already fail to be Borel for continuous real f .

Novelty checks were bounded as follows. The run indexes were searched for the source arXiv id, $D1$, non-Borel Dirichlet dilation, and simplicial function-space terms. The local source of the 2025 follow-up [2] was checked around its Dirichlet subsection; its results are vector-valued and assume a different “contains constants” setting. Web phrase searches for the exact Question 7.10 wording and the real non-Borel $D1$ variant did not reveal a later answer.

REFERENCES

- [1] O. F. K. Kalenda and J. Spurny, *On simpliciality of function spaces not containing constants*, arXiv:2401.16100.
- [2] O. F. K. Kalenda and J. Spurny, *Simpliciality of vector-valued function spaces*, arXiv:2501.12876.